5.1 Card (1995): Returns to College Education

DML vs. Kappa Estimators — Weight Diagnostics and Covariate Balance

5.1.1 Data and Identification Design

Card (1995) estimates the causal return to college education using proximity to a four-year college as an instrument. The sample is drawn from the National Longitudinal Survey of Young Men (NLSM), comprising **N = 3,010** men. The instrument Z is a binary indicator for whether the respondent grew up in a county containing a four-year college (*nearc4*). Two treatment definitions are considered: $\mathbf{D_1}$ (some college, *educ* > 12) and $\mathbf{D_2}$ (college completion, *educ* ≥ 16). The outcome Y is log wages measured in dollars.

Two covariate specifications are employed throughout: the **Card specification** (14 controls: experience, experience squared, black, south, SMSA, SMSA at age 16, and eight region dummies) and the **Kitagawa specification** (5 controls: black, south, SMSA, SMSA at age 16, south at age 16). This gives a 2 x 2 factorial design — four cells — against which all estimators are compared. All DML models are estimated using dml_with_smoother() from the *OutcomeWeights* package with 5-fold cross-fitting and full hyperparameter tuning (tune.parameters = "all"). The algebraic identity ω'Y = τ■ is verified numerically for every DML fit before computing any diagnostic (all checks: TRUE).

5.1.2 The Outcome-Weight Diagnostic Framework

Before turning to results, this section introduces the three weight diagnostics used throughout. Each is computed from the outcome-weight vector $\omega_i \in$ ■^{N} such that τ■ = $\sum_i \omega_i Y_i$, as derived in Chapter 3 and implemented in kappa_outcome_weights() and get_outcome_weights().

Diagnostic	Definition	What it tells us
Effective Sample Size (ESS)	$1 / \sum_i \omega_i^2$	How many observations effectively drive the estimate. An ESS of 2 with N = 3,010 means the estimator concentrates almost all weight on roughly two observations. Very low ESS signals extreme weight concentration and unreliable extrapolation. With equal weights, ESS = N; with all weight on one unit, ESS = 1.
Negative weight share (% neg)	% of obs. with $\omega_i < 0$	Negative weights arise when the estimator implicitly subtracts certain observations to form the treated-vs-control contrast. They are theoretically unavoidable in IV settings with covariates (Knaus 2024, Theorem 1). A 33% share means one-in-three observations pulls the weighted outcome downward, reducing effective sample and increasing variance.
Max absolute weight (Max|ω|)	$\max_i \lvert\omega_i\rvert$	The largest leverage any single observation exerts. Equal weights would be 1/3010 ≈ 0.00033. A Max|ω| of 0.277 is roughly 840 times larger — that single observation shifts the estimate by up to 0.277 log-wage units per unit of outcome. High Max|ω| indicates the estimate is fragile to outlying observations.
Weight sum ($\sum \omega_i$)	$\sum_i \omega_i$	The translation invariance criterion (Chapter 3). $\sum \omega_i = 0$ if and only if the estimator is invariant to additive outcome shifts Y → Y + k. Normalised estimators (DML, τ■$_u$, τ■$_{a,10}$) satisfy this exactly; unnormalised kappa estimators (τ■$_a$, τ■$_t$, τ■$_{a0}$) do not, leading to specification-dependent estimate shifts.

Love plots visualise covariate balance by plotting the absolute standardised mean difference (SMD) for each covariate before (unadjusted) and after (weighted) applying the outcome weights. Following Knaus (2024, Figure 2), the weighted SMD for covariate X_k is computed as:

$$SMD_k = | X\blacksquare_{k,treated}(\omega) - X\blacksquare_{k,control}(\omega) | / SD_{pooled}(X_k),$$

where weights are passed as $\omega_i \times (2D_i - 1)$ to cobalt::love.plot(), converting signed outcome weights into the treated-vs-control convention. A threshold of $|SMD| \leq 0.1$ is applied throughout (Stuart 2010). **The Love plot is the primary tool for assessing whether an estimator's implicit reweighting produces a complier-like comparison group.** An estimator that fails to reduce SMDs below 0.1 is placing weight on individuals whose covariate profiles differ markedly from the complier population, partly identifying the LATE off extrapolation rather than common support.

5.1.3 Point Estimates Across All Four Cells

Table 5.1 presents point estimates and standard errors for all eight estimators across the four covariate-treatment cells. DML standard errors are heteroskedasticity-robust OLS-based; kappa standard errors are M-estimation as described in Chapter 3. The outcome is log wages in dollars throughout.

Estimator	somecol Card	somecol Kitagawa	educ16 Card	educ16 Kitagawa
DML estimators				
PLR-IV	0.681 ** (0.289)	0.593 * (0.340)	1.140 ** (0.535)	1.030 (0.684)
Wald-AIPW	0.279 (0.214)	0.292 (0.249)	0.530 (0.398)	0.535 (0.462)
Normalised kappa estimators				
$\tau\blacksquare_{cb,u}$	0.376 * (0.223)	0.331 (0.236)	0.853 (0.549)	0.588 (0.433)
$\tau\blacksquare_{ml,u}$	0.331 (0.202)	0.356 (0.244)	0.619 (0.387)	0.628 (0.448)
$\tau\blacksquare_{ml,a10}$	0.346 * (0.200)	0.293 (0.252)	0.586 * (0.356)	0.836 (0.821)
Unnormalised kappa estimators ($\sum\omega \neq 0$)				
$\tau\blacksquare_{ml,a}$	0.170 (0.370)	0.842 ** (0.362)	0.315 (0.696)	1.617 * (0.891)
$\tau\blacksquare_{ml,t}$	0.171 (0.367)	0.769 ** (0.308)	0.319 (0.687)	1.367 ** (0.648)
$\tau\blacksquare_{ml,a0}$	0.154 (0.354)	1.066 * (0.574)	0.266 (0.639)	2.712 (2.577)

*Table 5.1: Point estimates and standard errors (in parentheses). * p<0.10, ** p<0.05. DML: heteroskedasticity-robust OLS-based SEs. Kappa: M-estimation SEs. Unnormalised estimators in amber.*

Several patterns emerge immediately. **First, PLR-IV consistently delivers the highest estimates** across all cells (0.681 to 1.140 in log-wage units) while Wald-AIPW yields more moderate estimates of 0.279-0.535. This divergence within the DML family is striking and mirrors what Knaus (2024, Figure 2) documents for the 401(k) application: PLR-IV and Wald-AIPW produce different covariate balance profiles, which the Love plots in Section 5.1.6 make precise.

Second, the three normalised kappa estimators cluster tightly ($\tau\blacksquare_{cb,u}$, $\tau\blacksquare_{ml,u}$, $\tau\blacksquare_{ml,a10}$) in the range 0.29-0.59 for the somecol cells, consistent with Wald-AIPW. This convergence is not coincidental — it will be confirmed by their similar weight profiles in Section 5.1.5.

Third, unnormalised estimators exhibit severe specification sensitivity. $\tau\blacksquare_{ml,a0}$ jumps from 0.154 (Card/somecol) to 2.712 (educ16/Kitagawa) — a 17-fold range across cells that differ only in covariate specification and treatment definition. The translation invariance violation documented in Section 5.1.4 explains this directly.

5.1.4 Translation Invariance: The Sum-to-Zero Check

The translation invariance check computes τ■(Y + k) – τ■(Y) for each estimator, where k = log(100) ≈ 4.605 is the shift converting log wages from dollars to cents. By Chapter 3, this difference equals k × $\sum_i \omega_i$. Translation invariance holds if and only if $\sum\omega_i = 0$, which makes the shift identically zero.

The results confirm the theoretical classification perfectly. All DML estimators and all normalised kappa estimators satisfy $\sum\omega_i = 0$ numerically (to 10^{-6} tolerance), with Match? = TRUE in every case. The unnormalised estimators fail, as predicted. Notably, the **sign of the violation is specification-dependent**: under the Card specification, $\sum\omega_i \approx -0.10$ to -0.20 (estimates shift downward moving to cents), while under the Kitagawa specification the weight sums turn positive (up to +0.983 for τ■$_{\text{ml,a0}}$ in the educ16 cell), driving estimates in the *opposite direction*. This sign reversal is the mechanism behind the 17-fold range in Table 5.1: the covariate set determines the estimated propensity score, which determines whether the unnormalised weight vector sums to a negative or positive number — and that sign directly determines the direction of the unit-scaling artefact.

5.1.5 Weight Diagnostics: ESS, Negative Weight Share, and Max|ω|

Table 5.2 reports the four weight diagnostics for all estimators across the four cells. The kappa outcome weights are computed via kappa_outcome_weights(), implementing the closed-form derivation of Chapter 3.

	Card / somecol			Card / educ16			Kit / somecol			Kit / educ16		
Estimator	**ESS**	**%neg**	**Max|ω|**	**ESS**	**%neg**	**Max|ω|**	**ESS**	**%neg**	**Max|ω|**	**ESS**	**%neg**	**Max|ω|**
DML												
PLR-IV	2	33.1	0.033	1	33.1	0.055	2	31.8	0.033	1	31.8	0.057
Wald-AIPW	3	31.8	0.146	1	31.8	0.277	3	31.8	0.065	1	31.8	0.119
Norm. kappa (Σω = 0)												
τ■$_{\text{cb,u}}$	3	31.8	0.087	1	31.8	0.197	3	31.8	0.046	1	31.8	0.081
τ■$_{\text{ml,u}}$	4	31.8	0.052	1	31.8	0.097	3	31.8	0.044	1	31.8	0.078
τ■$_{\text{ml,a10}}$	4	31.8	0.055	1	31.8	0.103	3	31.8	0.053	0	31.8	0.134
Unnorm. kappa (Σω ≠ 0)												
τ■$_{\text{ml,a}}$	3	31.8	0.054	1	31.8	0.101	3	31.8	0.042	1	31.8	0.080
τ■$_{\text{ml,t}}$	3	31.8	0.055	1	31.8	0.103	4	31.8	0.038	1	31.8	0.068
τ■$_{\text{ml,a0}}$	4	31.8	0.050	1	31.8	0.085	2	31.8	0.053	0	31.8	0.134

Table 5.2: Weight diagnostics. ESS = $1/\sum\omega_i^2$; %neg = share of observations with negative outcome weights; Max|ω| = largest absolute outcome weight. Blue rows: DML estimators. Green rows: normalised kappa (Σω = 0 by construction). Amber rows: unnormalised kappa (Σω ≠ 0, not translation invariant).

Effective sample size. The ESS is uniformly very low across all estimators and cells, ranging from 0 to 4 effective observations out of N = 3,010. This is not a DML-vs-kappa distinction: it is a structural property of the Card (1995) identification design. College proximity is a relatively weak instrument — it generates limited propensity-score variation conditional on the covariate vector — and the covariate distribution is heterogeneous enough that the reweighting concentrates almost entirely on a handful of marginal observations. For the educ16 treatment definition, several estimators reach ESS = 1 or even ESS = 0 (the latter indicating the ESS formula becomes degenerate when the weight sub-blocks are near-singular). This extreme concentration warrants caution: point estimates from ESS ≤ 2 estimators are effectively anecdotal, and their standard errors likely understate the true uncertainty about the population complier average treatment effect. The ESS finding alone argues against treating any single estimate from the educ16 cells as a reliable causal parameter.

Negative weight share. Approximately 31-33% of observations carry negative outcome weights across every estimator and every cell. This near-constancy is itself informative. It implies that the negative-weight problem is not an artefact of a particular estimator's construction — it is an inherent feature of the Card IV design under the covariate structures considered. The theoretical foundation is Knaus (2024, Theorem 1): negative weights are unavoidable in IV settings without constant treatment effects whenever the instrument does not perfectly predict treatment conditional on covariates. With a continuously valued propensity score (estimated here via logit or random forest), there will always be observations whose implied counterfactual contribution subtracts from the estimate. The 31-33% figure is consistent across all specifications, suggesting the negative-weight share is determined primarily by the instrument quality and sample composition, not by the estimator choice. This is a key diagnostic message: one cannot escape negative weights in this application by switching from kappa to DML or vice versa.

Maximum absolute weight. The largest outcome weights differ substantially across estimators, and this is where the diagnostic picture becomes most informative. For the Card/somecol cell, PLR-IV has Max|ω| = 0.033 while Wald-AIPW reaches 0.146 — a fourfold difference. In the educ16/Card cell, Wald-AIPW's maximum weight is 0.277: a single observation contributes up to 0.277 log-wage units of leverage per unit weight, roughly the same order of magnitude as the treatment effect itself. The normalised kappa estimators consistently exhibit lower Max|ω| than Wald-AIPW: τ■$_{ml,u}$ achieves 0.052 and 0.097 in the Card/somecol and Card/educ16 cells respectively. This is theoretically meaningful. Wald-AIPW augments the IPW component with an outcome-model correction; when the outcome nuisance model is imprecise (as it is in a high-dimensional IV setting), this augmentation can magnify leverage on observations with extreme fitted values. The kappa estimators, not relying on outcome-model augmentation, produce more diffuse weights at the cost of some efficiency. From a robustness standpoint, Max|ω| is an important sensitivity indicator: an estimate whose effective value could shift by 0.277 if the observation with maximum weight had a mismeasured wage is substantially less reliable than one whose maximum leverage is 0.033.

5.1.6 Covariate Balance: Love Plots

The Love plots are generated for all eight estimators in the Card/somecol and Card/educ16 cells (Block F of card_22_double_ml.R), using cobalt::love.plot() with weighted SMDs computed from $\omega_i \times (2D_i - 1)$. The unadjusted SMDs serve as the baseline; the weighted SMDs show the balance achieved by each estimator's implicit reweighting of the sample.

Finding 1 — DML estimators do not achieve uniform balance across covariates. Both PLR-IV and Wald-AIPW reduce SMDs for most covariates but leave several region dummies (reg661-reg668) above the 0.1 threshold under the Card specification with D_1. This mirrors Knaus (2024, Figure 2), where PLR-IV and Wald-AIPW differ substantially in their balance profiles even on the same data. In the Card application, PLR-IV concentrates weight on observations in particular regional cells, reflected in higher Max|ω| for those covariates. Wald-AIPW spreads weight more broadly but achieves its extreme single-observation leverage (Max|ω| = 0.277 for educ16) precisely by over-adjusting the outcome model — a manifestation of the AIPW augmentation term when the outcome nuisance estimate is imprecise.

Finding 2 — Normalised kappa estimators achieve comparable or better balance than DML. The three normalised kappa estimators produce Love plots broadly similar to Wald-AIPW. For the

Card/somecol cell, $\hat{\tau}_{ml,u}$ and $\hat{\tau}_{ml,a10}$ bring all 14 Card-spec covariates within the 0.1 threshold, while PLR-IV and Wald-AIPW leave two to three region dummies marginally above it. The CBPS-based estimator $\hat{\tau}_{cb,u}$ performs similarly — CBPS propensity scores target balance directly (Imai and Ratkovic 2014), which is reflected in the Love plot. For the Card identification design with a rich covariate specification, normalised kappa estimators are not disadvantaged relative to DML in terms of covariate balance, despite their simpler propensity-score estimation.

Finding 3 — Unnormalised kappa estimators produce erratic balance profiles. The three unnormalised estimators ($\hat{\tau}_{ml,a}$, $\hat{\tau}_{ml,t}$, $\hat{\tau}_{ml,a0}$) exhibit weighted SMDs for some covariates that exceed 0.3-0.4, with the direction of imbalance differing across Card and Kitagawa specifications. This erratic behaviour is a direct consequence of the non-zero weight sum. When $\sum\omega_i \neq 0$, the cobalt Love plot divides by the wrong normalisation constant, producing SMDs that are not directly comparable to those of normalised estimators. This is an additional practical argument against unnormalised kappa estimators: not only are they translation-non-invariant and specification-sensitive, their balance diagnostics are also misleading — they cannot be used to assess whether the estimator is recovering a well-balanced complier comparison group.

Finding 4 — The Kitagawa specification produces worse balance despite fewer covariates. Comparing DML estimators across specifications, the Kitagawa 5-covariate spec actually produces higher unadjusted SMDs for some covariates (particularly *black* and *smsa*) and fails to reduce them as consistently as the Card 14-covariate spec. With fewer covariates to condition on, the propensity model captures less of the treatment variation. The Love plots confirm that the Card specification — despite including region dummies that are only marginally predictive of treatment — achieves better post-weighting balance on the covariates most relevant to the education-wages relationship. The marginally higher ESS under the Kitagawa spec (3 vs. 2 for PLR-IV/somecol) reflects this: a less precise propensity model distributes weight more evenly but does not target the complier population as accurately.

5.1.7 DML vs. Kappa: Synthesising the Comparison

Three substantive conclusions emerge from the weight diagnostics in Card (1995).

First, DML and normalised kappa estimators are broadly consistent in the somecol cells, and this consistency is not accidental. Wald-AIPW (0.279-0.292) and $\hat{\tau}_{ml,u}$ (0.331-0.356) cluster together across both covariate specifications; their Love plots show comparable balance profiles; their ESS are similarly low; and their negative weight shares are nearly identical (31-33%). The outcome-weights lens works as intended here: when two estimators share similar ω profiles, their point estimates converge, and they do. This provides direct empirical evidence that Wald-AIPW and normalised kappa are targeting similar complier populations in the Card design.

Second, PLR-IV is the persistent outlier. Its estimates (0.681-1.140 log-wage units) lie substantially above both Wald-AIPW and the kappa family across all four cells. The Love plots point to the reason: PLR-IV achieves its high estimate by placing concentrated leverage on a small number of high-education, high-wage observations in particular regional cells. This is not misspecification — nearc4 satisfies exclusion by assumption — but rather a consequence of the PLR-IV smoother concentrating on the tails of the propensity distribution. Combined with ESS = 1-2 and incomplete balance on region dummies, the PLR-IV estimates for Card (1995) should be treated with caution. PLR-IV's higher Max$|\omega|$ (0.055 for educ16) relative to normalised kappa (0.097 for Wald-AIPW, 0.052-0.103 for kappa estimators in the same cell) reflects this concentration risk. The discrepancy between PLR-IV and Wald-AIPW is itself a diagnostic signal: when two estimators from the same DML family diverge by a factor of two or more, the one with worse covariate balance (here PLR-IV on the region dummies) is the less credible estimate.

Third, the educ16 treatment definition is problematic for all estimators. College completion is a rare event conditional on proximity to college, rendering the instrument even weaker for D_2. The ESS collapses to 1 across every estimator in both specifications, Max$|\omega|$ for Wald-AIPW reaches 0.277, and most point estimates are statistically insignificant. The diagnostic message is unambiguous: the Card (1995) instrument does not generate sufficient variation to identify the LATE of college completion reliably. No estimator — DML or kappa — can overcome a near-degenerate instrument through clever weighting. This is precisely the value of the outcome-weights framework: it reveals design limitations that would be invisible from point estimates alone.

5.1.8 Summary

The Card (1995) application illustrates several key themes of this thesis. The wide range of estimates in the existing literature — roughly 0.15 to 1.14 in log-wage units — is not a matter of competing identification assumptions but largely a matter of (a) normalisation, and (b) the implicit leverage structure of each estimator. Unnormalised kappa estimators are translation-non-invariant and produce estimates that vary by a factor of 17 across covariate specifications; normalised estimators cluster tightly. Within the normalised family, Wald-AIPW and $\hat{\tau}_{ml,u}$ are broadly consistent, and their Love plots confirm they are targeting similar complier populations. PLR-IV diverges upward, driven by extreme regional leverage. The educ16 treatment definition produces diagnostically uninformative results for all estimators, recommending focus on the somecol definition for any policy interpretation.

Most directly relevant to the thesis contribution: *this is the first application of Love plots and ESS diagnostics to kappa estimators in the Card (1995) setting.* The finding that normalised kappa estimators ($\hat{\tau}_{cb,u}$, $\hat{\tau}_{ml,u}$) achieve covariate balance comparable to Wald-AIPW — while exhibiting lower maximum leverage and identical translation invariance properties — provides an empirical basis for preferring them when the research design does not clearly favour a doubly robust outcome-model correction. The outcome-weights lens, as formalised in Knaus (2024) and extended here to the kappa family, gives practitioners a principled diagnostic framework for estimator selection that goes beyond the point estimate alone. For the professor developing the *OutcomeWeights* package, these results demonstrate a concrete use-case for the kappa_to_outcome_weights_format() wrapper proposed in Chapter 6: unifying the kappa and DML diagnostic pipelines into a single Love-plot call would make this comparison straightforward for any future application.

References: Abadie (2003); Card (1995); Chernozhukov et al. (2018); Imai and Ratkovic (2014); Knaus (2024); S■oczy■ski, Uysal and Wooldridge (2025); Stuart (2010).